SARAH MACGILLIVRAY

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SUMMARY

Adaptable systems engineer experienced in low-level firmware (C++), cloud orchestration (Python/FastAPI), and test automation. Proven ability to eliminate workflow bottlenecks, harden codebases, and deliver end-to-end solutions from hardware-adjacent code to high-level backend platforms.

SKILLS

Languages	C++, C, Python, Java, Javascript, SystemVerilog, ARM Assembly
Embedded & Hardware	FreeRTOS, ROS2, I2C/SPI, PCB Design (Altium), FPGA (Quartus/ModelSim), Oscilloscopes
Software & Systems	Linux Systems, Multithreaded Programming, Embedded Software Development, Client–Server Architecture
Developer Tools	Claude Code, Git/GitHub, Linux CLI, VS Code

EDUCATION

Bachelor of Applied Science, Computer Engineering – University of British Columbia | GPA: 87% **09/2022 – 04/2027 (expected)**

EXPERIENCE

TELUS – Software Development Engineering Co-op **05/2026 – 08/2026 | Vancouver, Canada**

- Built a Python/FastAPI app automating multi-system firmware deployments and device status campaigns across TR-069/USP, saving up to 20 engineering hours per month.
- Built automated CI-to-Jira test pipeline (Python, PowerShell, Xray API), saving 7 hrs/week with a pass/fail tracking dashboard.
- Presented platform build-vs-buy analysis to senior leaders, supporting a multi-million-dollar device management strategy.

GLUXKIND, INC. – Embedded Systems Engineering Co-op **05/2025 – 08/2025 | Vancouver, Canada**

- Architected a multithreaded C++ audio streaming engine on FreeRTOS, achieving sub-10ms latency for a therapeutic audio product supporting music libraries, albums, and playback controls.
- Built a mock BLE client web server to stress-test the system and support mobile app development.
- Engineered a fault-tolerant IPC system using FreeRTOS message queues to coordinate reliable audio delivery across three ESP32s and robust SD card interactions during concurrent system operation.
- Built a real-time serial diagnostic platform and optimized logging macros/dev tools, reducing internal debugging cycles by 75%.

DARKVISION – Electrical Engineering Co-op **05/2024 – 12/2024 | North Vancouver, Canada**

- Designed and implemented a noise-reduction amplifier PCB and SPI communication firmware that increased ultrasound signal strength by ~3× in harsh, attenuative fluids, expanding the range of sites where the product could be deployed.
- Programmed firmware for ultrasound hardware in C++/Python; collaborated via GitHub PR review to ship production code.
- Designed and produced PCBs (Altium) for space-efficient, reliable electronics in harsh downhole environments.
- Led R&D automation project building an HTML interface for PO/quoting, achieving a 90% reduction in quoting time.

TECHINSIGHTS – Image Processing Technician **05/2023 – 08/2023 | Ottawa, Canada**

- Trained ML algorithms for automated image correction and transistor extraction, cutting delivery timelines by multiple days.
- Analyzed semiconductor images at 4nm resolution (Apple A16 Bionic, Samsung, SK Hynix) to create virtualized circuit representations.
- Generated composite GDS-II files by stitching semiconductor layer images (vias, metals, polysilicon), facilitating comprehensive circuit documentation

TECHNICAL PROJECTS

ROS2 Autonomous Car: Developed LiDAR gap-following and steering algorithms for autonomous Jetson-based vehicles, handling complex obstacles and turns; integrated camera-based vision for supplemental guidance and achieved first place among 15 teams in both the obstacle course and standard racetrack. Then, implemented parallel and reverse parking in simulation and on the real track, using YOLOv8 for parking-sign detection and a Gemini API call for reliable OCR.

Autonomous Competition Robots: Designed and built competition-winning firefighter and sumo robots – custom PCBs, circuit schematics, and PIC40 microcontroller programming (Basic) for autonomous maze and sumo ring navigation.

AWARDS & ACTIVITIES

Awards: Dean's Honors List (2022–present), Academic All Canadian (2022–present), Women in Engineering Award (2023)

Activities: UBC Varsity Cross Country & Track Athlete, UBC Team Leadership Member (2022–Present)